



A SCALABLE MOVIE RECOMENDER SYSTEM

PRESENTER :

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BUSINESS PROBLEM

STREAMING PLATFORMS CONTAIN THOUSANDS OF MOVIES, MAKING IT DIFFICULT FOR USERS TO FIND CONTENT THEY WILL ENJOY.

- *USERS STRUGGLE TO DISCOVER RELEVANT MOVIES*
- *LOW ENGAGEMENT IF RECOMMENDATIONS ARE POOR*



STAKEHOLDERS

- ❖ STREAMING PLATFORMS
- ❖ MOVIE RECOMMENDATION SYSTEMS
- ❖ END USERS



OBJECTIVES

- ❖ Predict movie ratings for users
- ❖ Learn user preferences from past ratings
- ❖ Recommend Top 5 movies per user



DATASETS

SOURCE : GROUP LENS RESEARCH LAB

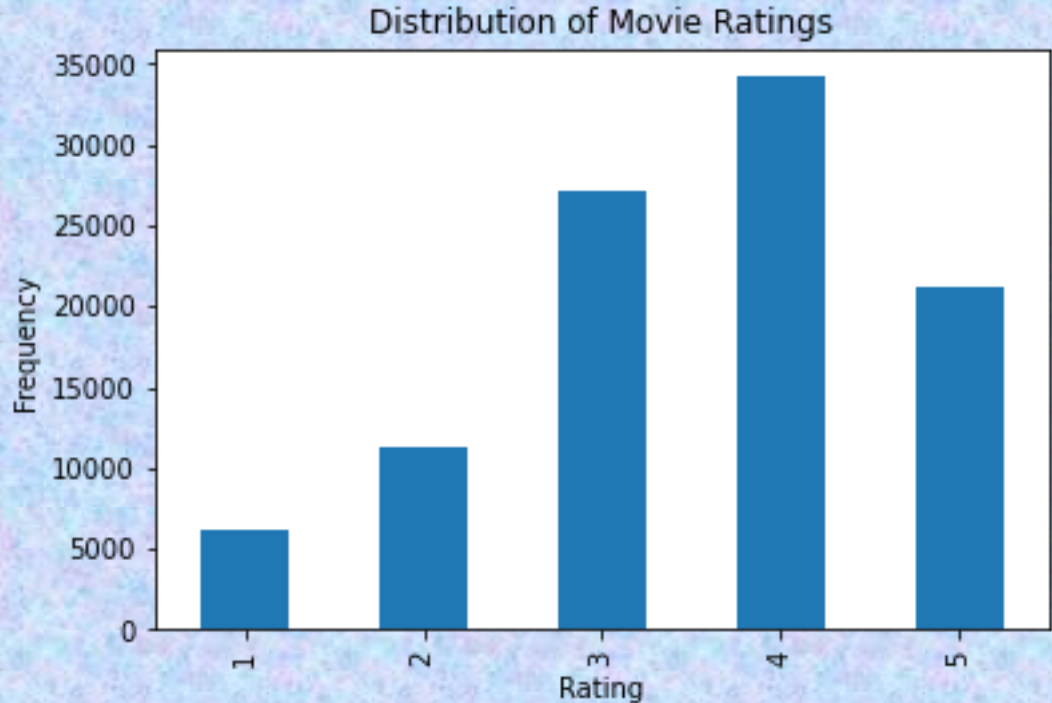
- ❖ RATING
- ❖ USERS
- ❖ MOVIES

KEY FEATURES:

- ❖ USERID
- ❖ MOVIEID
- ❖ RATING

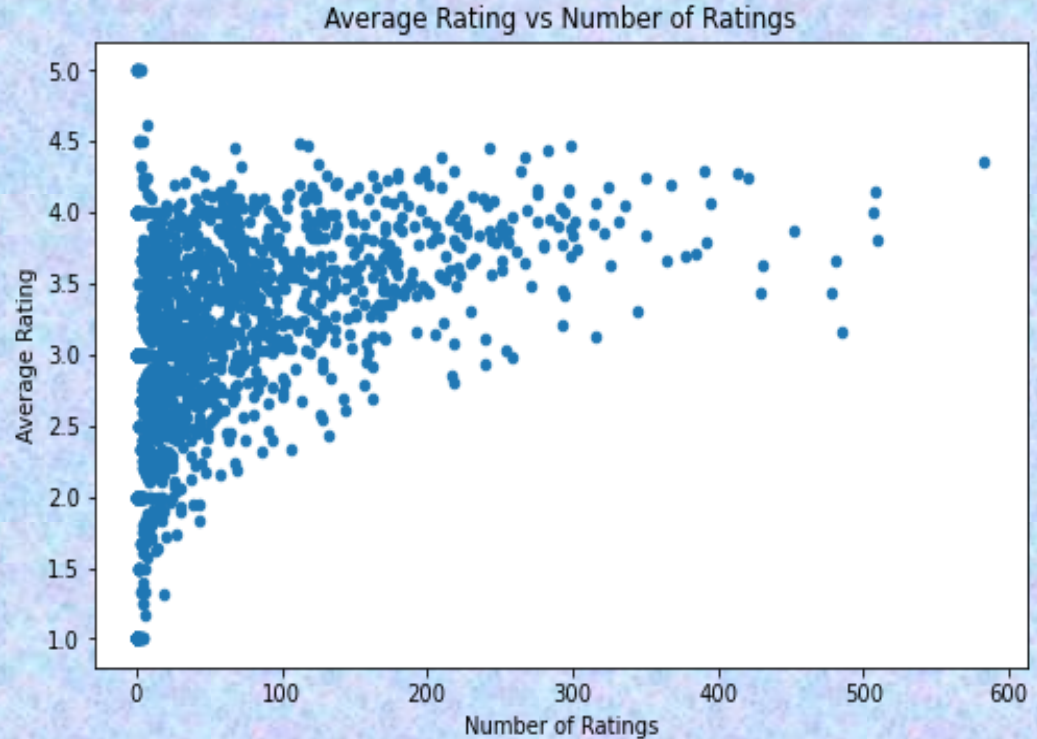
DISTRIBUTION OF RATING

- ❖ Bar chart showing frequency counts of rating from 1 to 5
- ❖ FINDING: skewed towards positive ratings
- ❖ Common score : 4 out of 5
- ✓ This shows that users tend to rate movies they already like



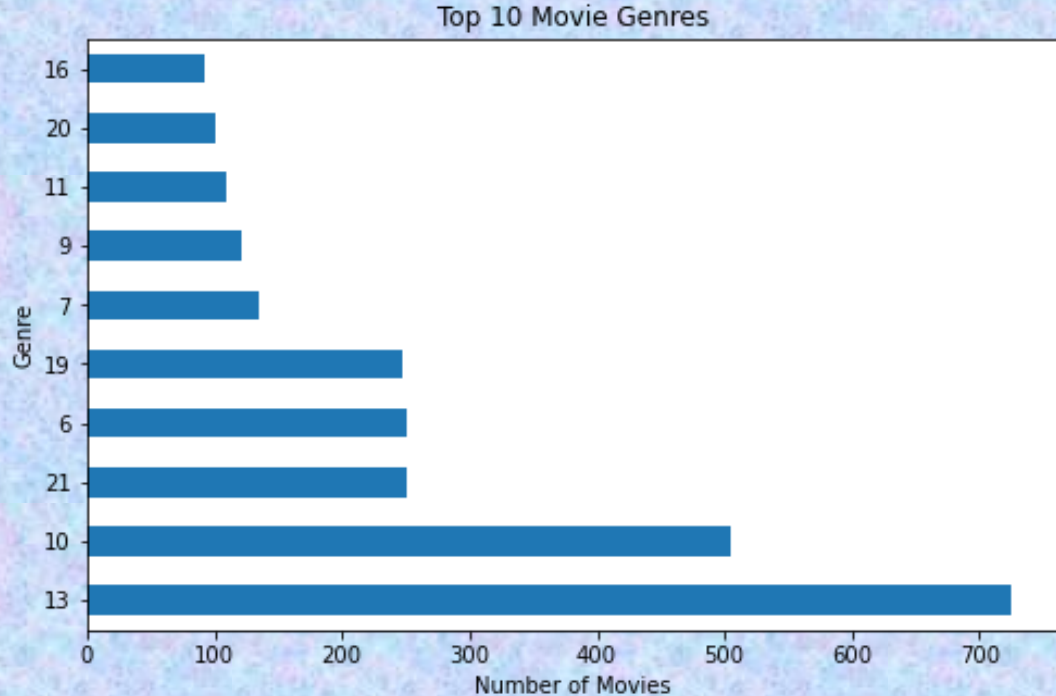
AVERAGE RATING VS MOVIE POPULARITY

- ❖ Scatter plot is tracking number of ratings against average rating value
- ❖ Focus: Bulk of movies have under 100 ratings
- ❖ Trend: highly rated movies have higher review volumes
- ✓ Business risk: high variance among rarely watched films



DISTRIBUTION OF MOVIE GENRE

- ❖ Horizontal bar chart displaying genre tag frequency
- ❖ Dominant classes: Drama, comedy and action
- ❖ Niche classes: documentaries and westerns presents the smallest foot print
- ✓ Svd value: latent factors map these unbalanced hidden traits easily

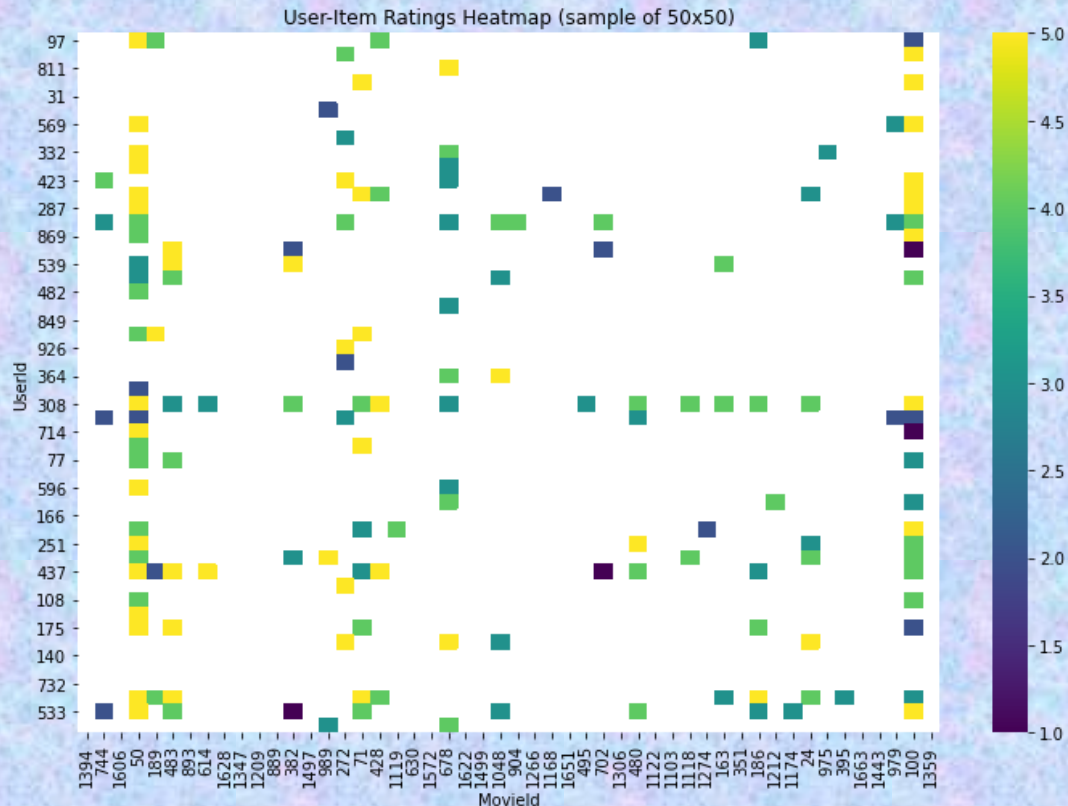




MODELING

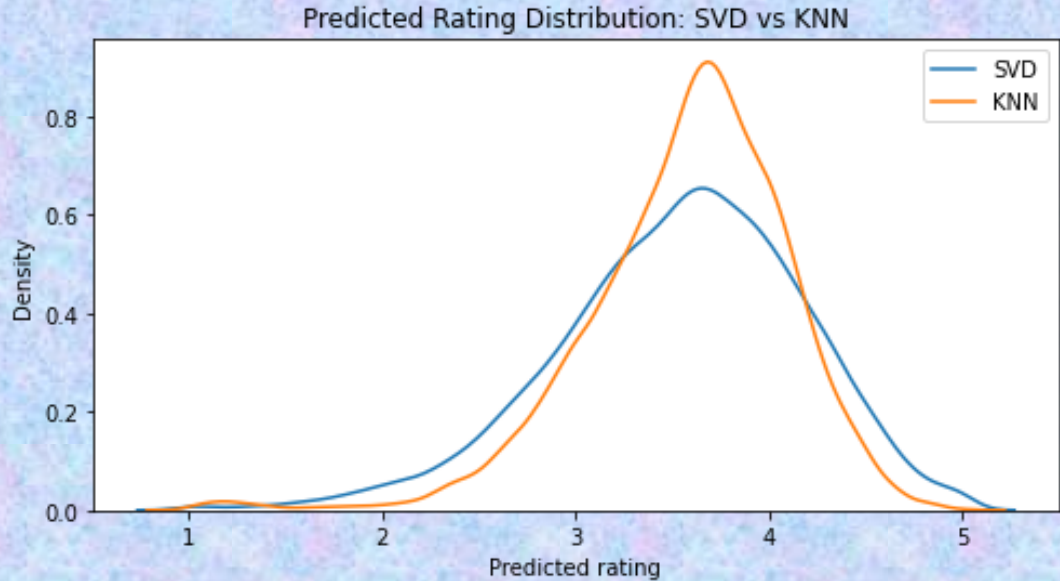
USER-ITEM HEATMAP

- ❖ Sparse nature of user item interaction motivates similarity-based approaches like Knn
- ❖ Top n recommendations: shows how SVD generates interpretable recommendations
- ❖ Distribution comparison: svd and knn prediction distribution differ



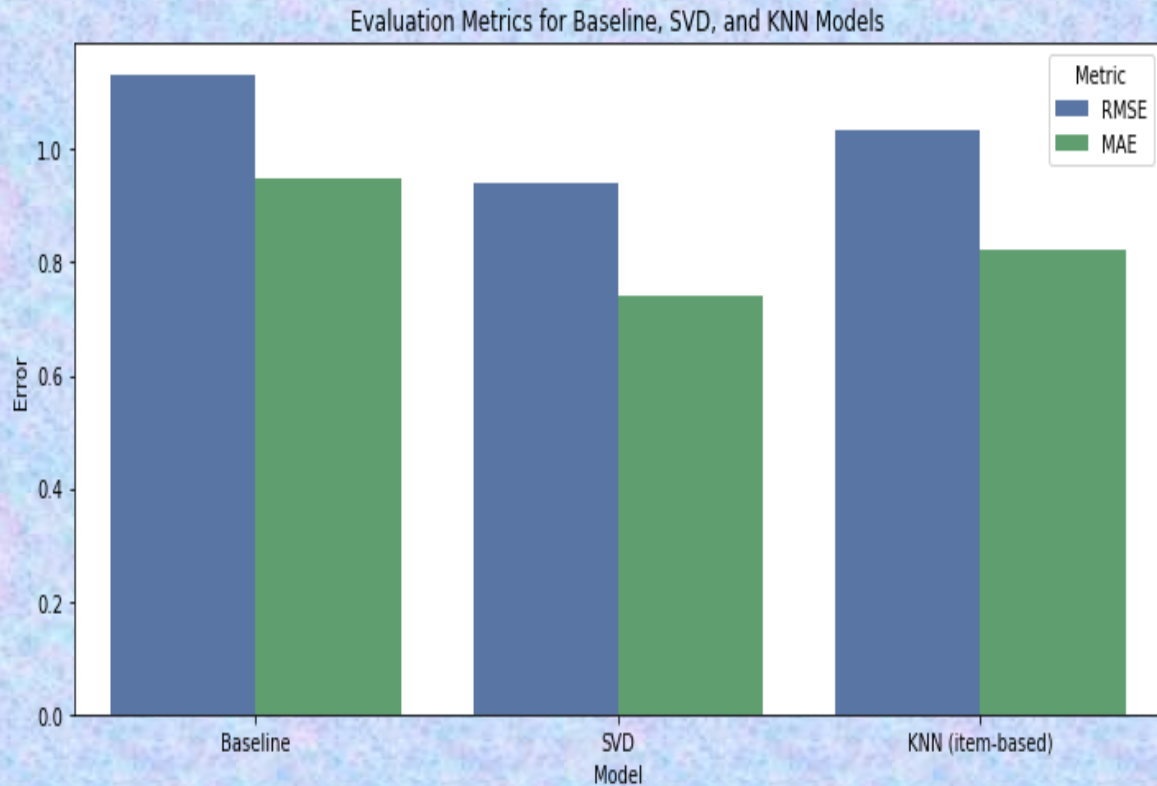
RATING DISTRIBUTION FOR SVD VS KNN

- ❖ Svd prediction: learned latent factor produce smoothly distributed factor
- ❖ Knn prediction: similarity-based approach may cluster predictions around users past rating
- ❖ Svd distribution reflects diverse user preferences



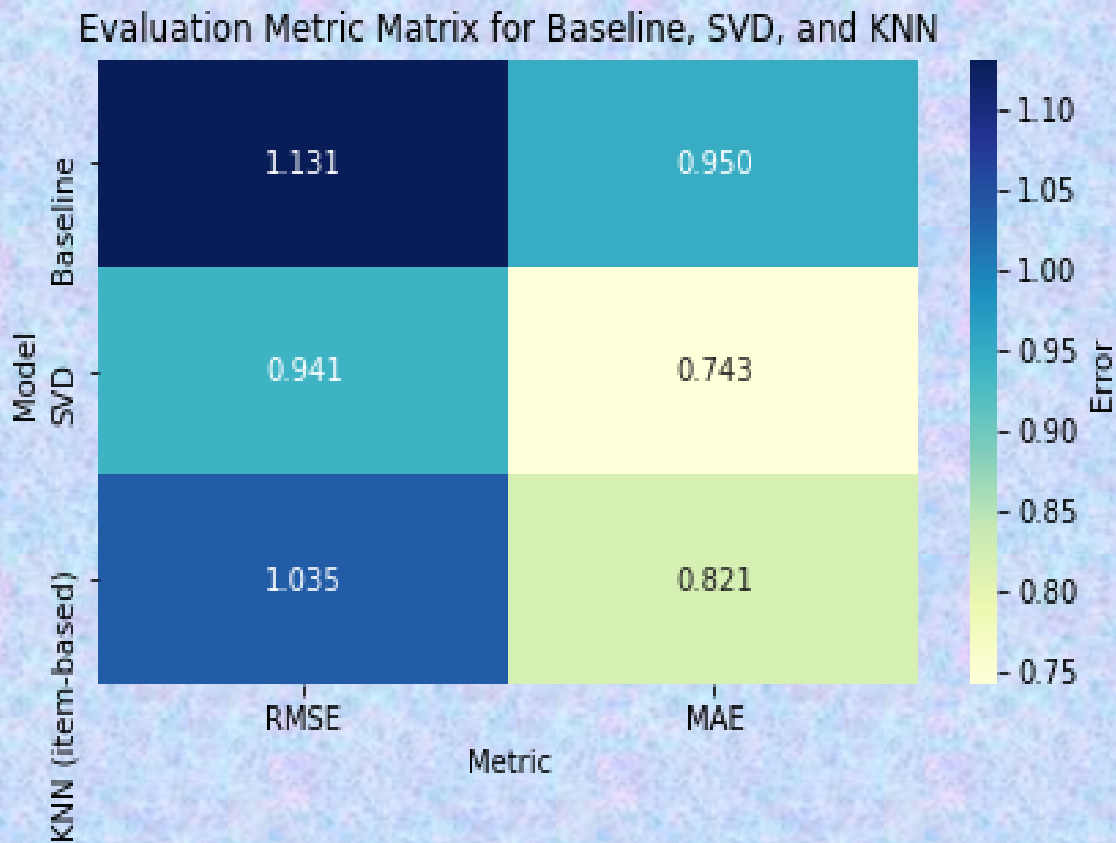
MODEL COMPARISON

- ❖ TO CHOOSE THE BEST MODEL WE CALCULATE THE AVERAGE ERROR (RMSE and MAE)
- ❖ MODEL WITH THE LOWEST AVERAGE ERROR BECOMES OUR CANDIDATE FOR DEPLOYMENT



MODEL SELECTION

❖ EVALUATION METRIC MATRIX FOR BASELINE, SVD AND KNN





ALIGNMENT WITH OBJECTIVES

- ❖ PERSONALISED RECOMMENDATIONS
- ❖ IMPROVED ACCURACY
- ❖ PRACTICAL DEPLOYMENT

WHY SVD IS THE BEST MODEL

- ❖ LOWEST ERROR RATE
- ❖ EFFICIENCY
- ❖ SCALABILITY



STAKEHOLDER BENEFITS

- **USER SATISFACTION**
- **B BUSINESS VALUE**
- **SCALABILITY**
- **INTERPRETABILITY**



FINAL INTERPRETATION

- Collaborative filtering improves recommendation quality over the baseline
- SVD and KNN are both evaluated as collaborative filtering options
- The selected best model is the strongest candidate for generating personalized movie recommendations



BUSINESS IMPLICATIONS

- Streaming platforms (Netflix-like systems)
- Movie websites
- Content recommendation engines



CONCLUSION

The three-model evaluation identifies the best performing model using RMSE and MAE.

This confirms that collaborative filtering improves prediction accuracy over the baseline.

The selected best model can now be used to generate personalized top 5 movie recommendations for users.



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